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DEPARTMENT OF ECONOMICS

Working Paper

A Unified Marxist Approach to Accumulation and Crisis in Capitalist Economies

by

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Working Paper 2017-21



**UNIVERSITY OF MASSACHUSETTS
AMHERST**

A Unified Marxist Approach to Accumulation and Crisis in Capitalist Economies

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Abstract

An economic crisis in capitalism is a deep and prolonged interruption of the economy-wide circuit of capital. Crises emerge from within the logic of capitalism's operation, and are manifestations of the inherently contradictory process of capital accumulation. The Marxist tradition conceptualizes two types of crisis tendencies in capitalism: a crisis of deficient surplus value and a crisis of excess surplus value. Two mechanisms that become important in crises of deficient surplus value are the rising organic composition of capital and the profit squeeze; two mechanisms that are salient in crisis of excess surplus value are problems of insufficient aggregate demand and increased financial fragility. This paper offers a synthetic and synoptic account of the Marxist literature on capitalist economic crises.

JEL Codes: B24; B51.

Key words: capitalism; crisis; rising organic composition; profit squeeze; underconsumption; financial fragility.

1. Introduction

For Marx, capitalism was an inherently crisis prone system of social production. In his account, crises emerged from within the very logic of capitalism and were a manifestation of the contradictions of the system. Using modern parlance, we can say that Marx conceived crisis as being endogenously generated by the functioning of capitalist systems. From this line of thinking comes the important conclusion that

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capitalism cannot exist without crises. Hence, any theory of the dynamics of capitalism must incorporate a theory of crisis as one of its integral components.¹

In adopting this viewpoint, Marx not only differed sharply from later-day neoclassical economists – who never tire of emphasizing the harmonious nature of capitalism – but also from major political economists of his time, including David Ricardo, who thought of crises as accidental phenomena, not related to the essential logic of capitalism. In modern parlance, Ricardo could probably be paraphrased as asserting that economic crises in capitalism were caused by exogenous shocks to the system.² Developing the logic of the endogenous conceptualization of crisis, and implicitly contrasting it with the diametrically opposed viewpoint that understands crisis as caused by exogenous factors, would provide a good entry point into Marxist analyses of capitalist crisis. But before we take that up, a small detour is in order.

To the extent we know on the basis of the extant literature, Marx did not leave behind a systematic analysis of capitalist crisis. This has been noted by later Marxist scholars, including Sweezy (1942), Foley (1986), and Heinrich (2012). While Marx (and Engels) referred to the phenomenon of capitalist crisis as early on in their political life as 1848 (in *The Communist Manifesto*), his comments on the topic are mostly fragmentary. In fact, they are not even collected together in one work but are instead scattered in various places in multiple texts. Many of these texts were unpublished during his lifetime like the *Notebooks of 1857-58* (the *Grundrisse*), the *Theories of Surplus Value* (written between 1861 and 1863) and Volumes 2 and 3 of *Capital* (written between 1863 and 1865). Hence, it seems clear that Marx never got the chance to return to these texts and fully work out his ideas on capitalist crisis.

Even though Marx's writings on capitalist crisis are incomplete and scattered, they can still be a source of important insights if approached with care. The key point to keep in mind when engaging with Marx's writings on capitalist crisis is that one can have a theory of capitalist crisis at two very different levels of abstraction. On the one hand, there can be a general theory of capitalist crisis at a very high level of

¹ Describing the raw power of capitalism, Marx noted how it "is destructive towards all [old ways of life], and constantly revolutionizes it, tearing down all the barriers which hem in the development of the forces of production, the expansion of needs, the all-sided development of production, and the exploitation and exchange of natural and mental forces." (Marx, 1993, pp. 410). But, continues Marx, this expansion and development of production is not a smooth process, it is beset with contradictions. The process of capital accumulation encounters internally generated limits. "But from the fact that capital posits every such limit as a barrier and hence gets *ideally* beyond it, does not by any means follow that it has *really* overcome it, and since every such barrier contradicts its character, its production moves in contradictions which are constantly overcome but just as constantly posited." (Marx, 1993, pp. 410).

² Like most classical economists, Ricardo also theorized about the eventual stationary state – a state of zero capital accumulation and growth – towards which capitalism was headed. For Ricardo, as for other classical economists, the stationary state was deeply problematic, a state of permanent crisis even. But Ricardo's theory about the evolution towards the stationary state rested on the growth of rent due to the expansion of agricultural production into lower quality land plots, in turn, caused by the growth of population. Thus, Ricardo's theory of crisis rested on factors external to the logic of capital accumulation. "Thus, economists like Ricardo, who take the capitalist mode of production as an absolute, feel here [i.e. with a decline in the average rate of profit] that this mode of production creates a barrier for itself and seek the source of this barrier not in production but rather in nature (in the theory of rent)." (Marx, 1991, pp. 350). In this sense, Ricardo's theory rested on exogenous factors, though not on exogenous *shocks* of the variety that is popular in contemporary macroeconomics.

abstraction. Such a theory tries to demonstrate, using political-economic reasoning, why crisis is built into the very logic of capitalism, why capitalism cannot be conceived without at the same time thinking about crisis. On the other hand, there can be theories of capitalist crisis at lower levels of abstraction. Such theories delineate specific economic mechanisms that can push a capitalist economy towards a crisis in a particular situation.

The main body, and most systematic part, of Marx's writings on capitalist crisis are of the first type; they are comments on a general theory of capitalist crisis, on the nature of economic crisis in capitalist economies understood at a very high level of abstraction. While one can find scattered comments on specific mechanisms that generate crisis in capitalism, with the most developed one being a discussion of the law of the tendential fall in the rate of profit in Volume three of *Capital*, these are mostly incomplete and unsystematic. It is only later scholars and activists who have picked up one or the other of Marx's comments on specific mechanisms and converted them into all-encompassing theories, or *the* theory, of capitalist crisis. Proponents of each of these theories have, then, spent inordinate amounts of time and energy in arguing why other theories are wrong or un-Marxian. One important task of this paper is to show that most of these controversies are unnecessary. A synthesis of Marxist theories of crisis can accommodate each of the important strands within one unified framework.

Before we present a unified treatment of Marxist theories of reproduction and crisis, at both high and low levels of abstraction, it might be useful to pay attention to some aspects of Marx's use of the term "crisis". At many places in his texts, Marx uses the term "crisis" to refer to what we would today call business cycle recessions, i.e. the downturn phases of regular business cycles. Marx often referred to these cycles as "industrial cycles" and the recessions as periods of "commercial crises". In these discussions, Marx highlighted the cyclical movements of credit and the rate of interest and noted the extreme scarcity of money during periods of crisis.³ But this is not the only way he used the term "crisis". In many instances, Marx used the term "crisis" in a deeper sense, as highlighting the inherent contradictions of the capital accumulation process, which, even if overcome temporarily, can never be fully surpassed. Used in this sense, the existence of "crises" point towards the historical relativity of the capitalist mode of production, towards the fact that it can be transcended, that it is not the natural way to organize the production of social wealth.⁴ In addition to these two different senses of the term

³ "If we consider the turnover cycles in which modern industry moves – inactivity, growing animation, prosperity, overproduction, crash, stagnation, inactivity, etc., cycles which fall outside the scope of our argument to analyze further – we find that a low level of interest generally corresponds to periods of prosperity or especially high profit, a rise interest comes between prosperity and its collapse, while maximum interest up to extreme usury corresponds to a period of crisis." (Marx, 1991, pp. 482).

⁴ Discussing the law of the tendential fall in the rate of profit, Marx notes: "The development of the productive forces of social labour is capital's historic mission and justification. For that very reason, it unwittingly creates the material conditions for a higher form of production. What disturbs Ricardo is the way that the rate of profit, which is the stimulus of capitalist production and both the condition for and the driving force in accumulation, is endangered by the development of production itself. And the quantitative relation is everything here. In actual fact, the underlying reason is something deeper, about which he has no more than a suspicion. What is visible in a purely economic manner, i.e. from a bourgeois standpoint of capitalist production itself, are its barriers, its relativity, the fact that it is not an absolute but only a historical model of production, corresponding to a specific and limited epoch in the development of the material conditions of production." (Marx, 1991, pp. 368).

“crisis”, Marx also sees crises as functional for capitalism, as corrective of underlying imbalances that are generated by the development of capitalism. The process of capital accumulation, according to Marx, generates contradictions, i.e. imbalances between relevant forces and tendencies, and crises are precisely the ways of restoring balance. But the restoration of balance is only temporary. The resolution of one contradiction creates conditions for the emergence of another, and so the development of capitalism proceeds through an endless series of contradictions.⁵

The rest of the paper is organized as follows. In section 2, I discuss a general theory of capitalist crisis based on Marx’s writings. In sections 3 and 4, I discuss in greater detail specific mechanisms that generate crisis tendencies in capitalist economies, drawing on the writings of Marx and later Marxist scholars. To organize the discussion of specific mechanisms identified by the Marxist tradition as causes of crisis in capitalism, I offer a typology in terms of two broad types of crises tendencies: a crisis of deficient surplus value, and a crisis of excess surplus value. In section 3, I discuss the first type of crisis – the crisis of deficient surplus value; and in section 4, I discuss the second type of crisis – the crisis of excess surplus value. I conclude the discussion in section 5 with some comments about important controversies in the Marxist literature on capitalist crisis. Some of the arguments advanced at various points in the paper can be stated and proved more precisely using basic mathematic formalism. All such mathematical proofs are collected together in the appendix.

2. The Nature of Capitalist Crises

2.1 Definitions

A convenient starting point for a Marxist analysis of crises in capitalism is the economy-wide circuit of capital, $M \rightarrow (P) \rightarrow C' - M'$, which is an abstract representation of the flow of value in a capitalist firm, or the whole capitalist economy, over time. One can conceptualize the capitalist economy as starting with a sum of money, M , and using it to purchase commodities, C , which includes both means of production and labour-power. The two are then combined in the process of production, (P) , with the output being the flow of finished commodities, C' . The circuit completes itself when the finished commodities are sold in the market for a sum of money, M' .

The difference between M' and M is surplus value, which is generated in production and realized through sale. The generation and realization of surplus value are the two phases of the circuit of capital, and together they constitute the primary motivations and the key drivers of the capitalist economy. As long as this process proceeds smoothly, as long as the unity of the two phases of the circuit of capital is intact, as long as the ‘normal’ amount of surplus value is generated in each cycle, a capitalist economy is

⁵ Discussing contradictory influences on the process of capital accumulation, Marx notes: “These various influences sometimes tend to exhibit themselves side by side, spatially; at other times one after the other, temporally; and at certain points the conflict of contending agencies breaks through in crises. Crises are never more than momentary, violent solutions for the existing contradictions, violent eruptions that re-establish the disturbed balance for the time being.” (Marx, 1991, pp. 356). But the balance is restored only for the time being, the barriers to accumulation are overcome only temporarily. “Capitalist production constantly strives to overcome these immanent barriers, but it overcomes them only by means that set up the barriers afresh and on a more powerful scale.” (Marx, 1991, 358).

in good health. But when there is large scale and long lasting interruption of this process, a capitalist economy enters a period of structural crisis. Let us note this as

Definition 1. A structural crisis of capitalism is a deep and prolonged interruption of the economy-wide circuit of capital.

Since a structural crisis of capitalism is a rupture of the economy-wide circuit of capital, it always manifests itself, first and foremost, as a *crisis of overproduction*, i.e. an accumulation of unsold and unsalable commodities. The rupture starts in the last phase of the circuit of capital, i.e. in the stage represented by C'-M', where the finished commodities are brought to the market to be sold. Hence, the first manifestation of crisis is the accumulation of a stock of unsold and unsalable commodities, but it quickly develops into a situation marked by the paradoxical coexistence of unused productive capacity and unmet human needs.⁶ To understand why the capitalist system gets caught in crises of overproduction ever so often, Marxist political economy offers insights at two levels of abstractions, a general theory of crisis at a high level of abstraction, and specific mechanisms of crisis at lower levels of abstraction.

2.2. A General Theory of Crisis

Marx's general theory of capitalist crisis is developed in the most systematic manner in Chapter 17 of Book II of the *Theories of Surplus Value* (Marx, 2000) and in Section Two of The Chapter on *Capital* in the *Grundrisse* (Marx, 1993). A most striking passage in the *Grundrisse* summarizes Marx overall understanding:

The whole dispute as to whether *overproduction* is possible and necessary in capitalist production revolves around the point whether the process of realization of capital within production directly posits its realization in circulation; whether its realization posited in the *production process* is its real realization. Ricardo himself, of course, has a suspicion that the exchange value of a commodity is not a value apart from exchange, and that it proves itself as a value only in exchange; but he regards the barriers which production thereby encounters as accidental, as barriers which are overcome. He therefore conceives the overcoming of such barriers as being in the essence of capital, although he often becomes absurd in the exposition of that view; while Sismondi, by contrast, emphasizes not only the encounter with the barriers, but their creation by capital itself, and has a vague intuition that they must lead to its breakdown. He therefore wants to put up barriers to production, from the outside, through custom, law, etc., which of course, as merely external and artificial barriers, would necessarily be demolished by capital. On the other side, Ricardo and his entire school never understood the really *modern crises*, in which the contradiction of capital discharges itself in great thunderstorms which increasingly threaten it as the foundation of society and of production itself. (Marx, 1993, pp. 410-411; emphasis as in original).

⁶ In earlier modes of production, crises were caused by the destruction of productive capacity through, for instance, natural disasters or wars, and manifested themselves as a *crisis of underproduction*. Thus, crisis was the expression of the fact that too little was produced. In capitalism, on the contrary, crisis is an expression of the fact that too much has been produced, which is manifested as unsold commodities.

In the text surrounding this passage and at other places in the *Theories of Surplus Value*, we find a detailed critique of the acceptance by orthodox economics of Say's Law, the proposition that generalized overproduction is not possible in capitalism.⁷ In modern parlance, this is the proposition that aggregate demand can never fall short of aggregate supply. Marx is critical of one-sided arguments of both supply-siders like Ricardo and demand-siders like Sismondi. He offers a more holistic approach and argues that the denial of the possibility of overproduction – the key characteristic of capitalist crisis – rests on two conceptual fallacies: ignoring money and abstracting from the specificity of capitalism. This argument has been developed in Sweezy (1942, chapter VIII) and Foley (1986, chapter 9), and I will summarize the main points here.

2.2.1. Simple Circulation of Commodities

In a barter economy, the possibility of rupture of the process of circulation (of the goods and services produced) is minimal. This is because every sale is immediately also a purchase. Thus, there is very little chance, once a barter economy has stabilized itself in terms of demand and supply, for the emergence of overproduction on a large scale.⁸ As soon as we move beyond barter and allow for money to mediate the process of exchange of goods and services, the possibility of a rupture of the circulation opens up. With money mediating exchange of commodities, the coincidence of sale and purchase is broken, and so sale and purchase can be separated – both in time and in space. Moreover, for Marx money emerges from within the very logic of commodity exchange (see Chapter 1-3, Volume One of *Capital*). Hence, it is conceptually impossible to ignore money when discussing a society where production is organized through exchange, i.e. every commodity producing system is always a monetary economy.

Armed with this insight let us turn to a study of what Marx calls the 'simple circulation of commodities': C-M-C'. Here, the producer of the commodity C sells it for a sum of money M and uses it to purchase another commodity C'. Since we are considering a commodity producing system, money necessarily mediates circulation of the commodities C and C'. Hence, there is a hidden possibility that the process of circulation will not be completed. The seller of C might not use the proceeds from the sale to immediately purchase C'. This means that the owner of C' will not be able to sell her commodity. But if the owner of C' is unable to sell her commodity, she will not buy some other commodity C'', and so on. This possibility immediately negates Say's Law: demand can indeed fall short of supply over some given period of time for a whole range of commodities. If this occurs on a large enough scale, generalized overproduction can emerge, i.e., aggregate demand can fall significantly short of aggregate supply at the level of the whole economy. This shows that Say's Law does not hold in commodity producing systems. Since capitalism is a specific form of a commodity producing system, where labour-power has also become a commodity, this critique is enough to demonstrate the *possibility* of a crisis of overproduction in capitalism. Even while this argument is formally true, it is difficult to understand why a rupture of the circulation process, of the kind elaborated above, would occur in any economy dominated by simple

⁷ "The attempts made from the orthodox economic viewpoint to deny that there is *general overproduction* at any given moment are indeed childish." (Marx, 2000, pp. 411). For an interesting discussion of different interpretations of "Say's Law", see Rotta (2017).

⁸ If there is a non-produced asset like land that is in circulation, then even a barter economy might face a problem of overproduction of commodities. For instance, if there is excess demand for land, that will imply, by Walras' Law, an excess supply of commodities (Rakshit, 1989).

commodity production. After all, there is a basic continuity that is demanded by consumption, and since the goal of production is largely determined by the needs of consumption in a simple commodity production system, it is unlikely that circulation would be ruptured on a large scale.

A whole new logic comes to the fore when we start considering a capitalist economy because here the typical form of circulation is not C-M-C' but rather what Marx calls the 'circulation of money', i.e. M-C-M'. Since surplus value can only be generated in production, hidden behind the circulation of money, at the aggregate level, is nothing but the circuit of capital, M-C -- (P) -- C'-M', which we have already encountered above. Here, we need to differentiate two important phases represented within the circuit of capital: the generation of surplus value during production, represented by (P), and the realization of surplus value through sale, represented by C'-M'. The circuit of capital is a unity of these two phases, and a structural crisis, by interrupting the flow of value through the circuit, breaks this unity. Since all exchanges represented in the circuit are monetary exchanges, this by itself opens up the possibility of interruption, as we have already noted. But there is an additional and important reason that relates to the specifically capitalist nature of the process.

2.2.2. The Specificity of Capitalism

The key driver of the circuit of capital is the generation and realization of surplus value in each cycle. But capitalist firms are interested not so much in the absolute amount of surplus value as they are in the amount of surplus value in relation to the amount of capital they advanced to begin the circuit. The ratio of surplus value and the capital advanced is the rate of profit. Thus, the key motivation and driver of the capitalist system is the need to continuously increase the rate of profit, and absent an increase, certainly to prevent it from falling. This gives us the proximate cause of crisis in capitalist economies: a decline in the rate of profit.

At any point in time, it is possible to define a threshold value for the average rate of profit that is deemed necessary for investment by capitalist firms (Sweezy, 1942). This threshold value can be understood as the minimum rate of return that justifies the 'trouble' of undertaking investment – the uncertainty and risk of future returns. When the average rate of profit falls below this threshold, capitalist firms drastically reduce investments or even stop investing altogether. If the reduction in capital outlays is large in magnitude and affects significant portions of the capitalist economy (or even significantly impacts key sectors with strong backward and forward linkages), it will lead to an immediate fall in aggregate demand in the whole economy. The reduction in capital outlays will also entail laying off currently employed workers or drastic reductions in hiring of new workers, which, in either case, will imply a fall in wage incomes. The fall in wage incomes will lead to a reduction in consumption expenditure by working class households and cause a further fall in aggregate demand, worsening the initial problem. If capitalist firms react to the decline in aggregate demand (and the emergence of excess capacity in key sectors) with a second round of reductions in capital outlays, this could very well be the beginnings of a deep and prolonged interruption of the economy-wide circuit of capital.

Definition 2. The proximate cause of crisis in capitalism is a fall in the average rate of profit.

The rate of profit is the ratio of the surplus value realized through sale of the commodity and the capital advanced to produce it. Hence there are two different, and mutually exclusive, ways in which the rate of profit can fall, which offer a typology of crisis tendencies in capitalism.

The first scenario is marked by a chronic insufficiency of demand in the market, so that the commodity is sold at a price that is below its value (or price of production). Hence, the sale of the commodity does not realize the full surplus value (or the average rate of profit). Thus, the *realized* rate of profit falls below the ‘normal’ rate of profit (which prevailed previously). This scenario is identified in this paper with a ‘crisis of excess surplus value’ (because more surplus value was produced than could be realized through sale).

In the second scenario, the commodity is sold at its full value (or price of production) and yet the realized rate of profit declines. Thus, in this case, the problem is not one of realization of the surplus value embedded in commodities, but rather points to the production of insufficient surplus value. Even when all the surplus value generated in production is realized through sale, the rate of profit still falls. This scenario is identified in this paper with a ‘crisis of deficient surplus value’ (because the system produces less surplus value than is necessary to ensure a normal rate of profit).⁹

In the next two sections of the paper, I turn to a detailed discussion of each of these two types of crisis – crisis of deficient surplus value and crisis of excess surplus value – focusing in particular on the key mechanisms that become important in each of them.

3. Crisis of Deficient Surplus Value

In a crisis of deficient surplus, the rupture in the economy-wide circuit is proximately caused by a fall in the rate of profit, which, in turn, is caused not by a fall in aggregate demand, but rests, instead, on two different mechanisms: (a) the rising organic composition of capital (Marx, 1991; Mattick, 1981; Shaikh, 1978; Kliman, 2011), and; (b) the profit squeeze (Marx, 1990; Glyn and Sutcliffe, 1972). The easiest way to grasp the logic of these two mechanisms is to start with the definition of the rate of profit and decompose it in terms of the rate of exploitation and the organic composition of capital.

3.1. The Rate of Profit

Let C and V represent constant capital and variable capital advanced, and S represent the surplus value generated, in the production of commodities.¹⁰ The value of commodities, W , is given by the sum of the three: $W = C + V + S$. We can define two ratios with respect to the production process:

⁹ This typology is used in Foley (2012). It is also implicit in Sweezy (1942, chapter VIII) even though he uses a different terminology. What I have termed crises of deficient surplus value is referred to by Sweezy (1942) as ‘crises associated with the falling tendency of the rate of profit’; what I have called crises of excess surplus value is termed by Sweezy (1942) as ‘realization crises’.

¹⁰ In his analysis, Marx mostly works with what we would now call a circulating capital model, and I follow him in that respect. In such a setting, there is no fixed capital and so all the means of production are used up in one cycle

$$Q = \frac{c}{v} \quad (1)$$

which represents the organic composition of capital, and

$$e = \frac{s}{v} \quad (2)$$

which represents the rate of exploitation. The rate of profit, r , is defined as the ratio of surplus value and total capital advanced (sum of constant capital and variable capital), and can be expressed in terms of the rate of exploitation and the organic composition of capital as follows¹¹:

$$r = \frac{s}{c+v} = \frac{\left(\frac{s}{v}\right)}{\left(\frac{c}{v}\right)+1} = \frac{e}{1+Q} \quad (3)$$

Equation (3) shows that the rate of profit will fall if: (a) the rate of exploitation falls, with the organic composition of capital remaining unchanged; and (b) the organic composition of capital rises, with the rate of exploitation remaining unchanged. These two routes for the possible fall in the rate of profit provide primary motivations for the two mechanisms that can lead to crises of deficient surplus value.

The profit squeeze mechanism focuses attention on situations where changes in the capitalist economy lead to a fall in the rate of exploitation. This leads to a fall in the numerator in equation (3), and under the assumption that the organic composition of capital does not fall too much, the rate of profit falls. On the other hand, the rising organic composition of capital mechanism directs attention to situations when forces in the capitalist economy push the organic composition of capital to rise, so that the denominator in equation (3) rises. Under the assumption that the rate of exploitation does not rise too much, the increase in the organic composition of capital leads to a fall in the rate of profit.

An immediate issue to consider is the nature and strength of the possible relationship between the two variables appearing in the definition of the rate of profit in (3), viz. the possibility of a functional relationship between the rate of exploitation and the organic composition of capital. The validity of both the profit squeeze mechanism and the rising organic composition of capital mechanism rest on the specific nature of such a relationship, if it exists at all, and one must make that explicit to develop a complete analysis of these two mechanisms. Intuition suggests that there is likely to be causal effects running in both directions, and, more importantly, that the relationship can be either positive or negative. Let us see why.

of production. In a more general setting with fixed capital, which Marx uses at times in Volume Three of Capital, the “constant capital”, C , would refer to the value of means of production *used up* in one cycle of production.

¹¹ The rate of profit, r , is defined as the ratio of the *flow* of surplus value over a period, s , to the *stock* of capital tied up in the circuit of capital, K , i.e. $r = s/K$. We can decompose the rate of profit as follows: $r = (S/V) * (V/(C + V)) * ((C + V)/K)$. In this decomposition, $C + V$ is the capital advanced and K represents the total capital tied up in the circuit of capital, so that the ratio of the two is the “rate of turnover of capital”, i.e. the number of production cycles over which the capital advanced exhausts the capital tied up. When the rate of turnover of capital is 1, we are dealing with the special case of a circulating capital model, where the capital advanced completely exhausts the capital tied up within one production cycle (as Marx does in much of his analysis). In that case, the rate of profit is given by the expression in equation (3) in the text.

When we consider a causal effect running from changes in the rate of exploitation to the organic composition, the relationship is likely to be negative. For instance, when changes in labour market conditions, like the depletion of the reserve army of labour or rapid unionization, increase the bargaining power of workers vis-a-vis capitalists, it is likely to translate into upward pressure on real wages, implying a fall in the rate of exploitation. Capitalists would respond with, among other things, the active search for and adoption of new techniques of production that save on the costly input, i.e. labour-power. Adoption of such labour saving technical change can, under some conditions, lead to an increase in the organic composition. Hence, in this case, we would expect a negative relationship between the rate of exploitation and the organic composition of capital.

On the other hand, when we consider a causal effect running in the other direction, i.e. from the organic composition to the rate of exploitation, the relationship is likely to be positive. This is because an increase in the organic composition is likely to reflect the increasing mechanization of the production process. The adoption of such capital-intensive techniques of production is likely to increase the productivity of labour, and if real wages do not move up one for one, the rate of exploitation will increase.

Thus, if the rate of exploitation is the initiator of change, the functional relationship between the organic composition and the rate of exploitation is likely to be negative, i.e. if there is an *exogenous* fall in the rate of exploitation, the organic composition of capital will, as a result, rise. On the other hand, if the organic composition is the origin of change, then the functional relationship between the rate of exploitation and the organic composition is likely to be positive, i.e. if there is an *exogenous* rise in the organic composition of capital, the rate of exploitation will, as a result, rise.

3.2. Profit Squeeze

The profit squeeze mechanism was elaborated by Marx in chapter 25 of Volume One of *Capital* (Marx, 1990), and referred to again in chapter 20 of Volume Two of *Capital* (Marx, 1992) and in chapter 15 of Volume Three of *Capital* (Marx, 1991). While it has been used by Dobb (1945) as a general theory of economic crisis in capitalism, it was also used by many Marxist scholars to offer an explanation of the crisis of the 1970s (see, for instance, Glyn and Sutcliffe, 1972; Bowles, et al. 1983). The underlying logic of the argument is straightforward. As capitalist economies emerge from business cycle recessions, capital outlays by capitalist firms pick up. With the growth in capital outlays comes the growth in the demand for labour-power. If the pace of capital outlays and capital accumulation remains high for several years, the labour market starts tightening and the reserve army of labour – what Marx calls the relative surplus population – starts depleting. As the reserve army becomes smaller, the bargaining power of workers increases, reflecting the decline in the ‘cost of job loss’: if a worker is laid off, she can find alternative employment relatively easily. The increase in the bargaining power of workers translates into higher real wages. Glyn and Sutcliffe (1972) argue, for the case of the British economy, that international competition constrained the ability of firms to raise prices and pass off the increased costs – due to rising real wages – to consumers. Thus, at some point, the growth rate of real wages outpaces the growth rate of labour productivity, and profits are squeezed. The result is a fall in the rate of

exploitation – the ratio of surplus value (profit) and variable capital (wages) – and finally a fall in the rate of profit, as can be seen from (3). The fall in the rate of profit chokes off capital outlays, leading to a crisis.

While theorist who used, or still use, the profit squeeze mechanism for explaining capitalist crisis did (does) not pay much attention to the possible effect of the fall in the rate of exploitation on the organic composition of capital, doing so will only strengthen the argument. If the rate of exploitation falls, capitalist firms will have a strong incentive to replace workers with machines. If successful, that will increase the organic composition of capital. Using (3), we can see that the rise in the organic composition of capital, as a response to the fall in the rate of exploitation, will depress the rate of profit further. Hence, if we allow for the possible feedback effect of the profit squeeze on the organic composition of capital, the overall decline in the rate of profit due to an exogenous fall in the rate of exploitation will be larger. This might contribute to a steeper fall in capital outlays and lead to a deeper crisis.

While Glyn and Sutcliffe (1972) have used it to offer an explanation of the crisis of the 1970s, it is not clear that Marx meant to use it for anything other than an explanation of the fluctuating pace of capital accumulation, what we would today call the ‘business cycles’. The account of the profit squeeze mechanism in Chapter 25 of Volume One certainly seems to be a theory of what modern economics calls business cycle fluctuations. Since structural crises of capitalism are radically different from normal business cycle recessions, it is not clear that the profit squeeze mechanism – as developed by Marx in Chapter 25 of Volume One – can, or was even meant to, offer a theory of structural crises in capitalism.

3.3. Rising Organic Composition of Capital

Among the various specific crisis mechanisms, the rising organic composition of capital mechanism was the one that was most systematically analysed by Marx. In chapter 13 of Volume Three of *Capital*, Marx discussed this mechanism under the title of the ‘law of the tendential fall in the rate of profit’ (Marx, 1991).

The starting point of Marx’s argument is the recognition of an important characteristic of capitalist production, viz. it’s growing mechanization, whereby the same quantity of labour-power works with a growing mass of raw materials and machinery. Hence, the volume of constant capital advanced, i.e. the money used to purchase the non-labour inputs into production, rises with respect to the volume of variable capital advanced, i.e. the money used to purchase labour-power. The result is an increase in the organic composition of capital with the development of capitalist production. If the rate of surplus value remains unchanged, the rising organic composition of capital will get expressed in the tendency for the average rate of profit to fall.

Since the mass of living labour applied continuously declines in relation to the mass of objectified labour that it sets in motion, i.e. the productively consumed means of production, the part of this living labour that is unpaid and objectified in surplus-value must also stand in an ever-decreasing ratio to the value of the total capital applied. But this ratio between the mass of

surplus-value and the total capital applied in fact constitutes the rate of profit, which must therefore steadily fall. (Marx, 1991, pp. 319)

There are at least two possible questions that arise with respect to this celebrated argument of Marx.¹² First: can we justify the assumption that the rate of exploitation remains unchanged even as the organic composition of capital increases exogenously? Second: can we offer a convincing explanation of why the organic composition of capital *must* rise over time?

3.3.1. Responsiveness of the Rate of Exploitation to the Organic Composition

The answer to the first question has already been discussed above: we *cannot* justify the claim that the rate of exploitation remains unchanged when the organic composition of capital increases. There is sound economic reasoning to suggest, as we have argued above, that exogenous increases in the organic composition increases the rate of exploitation. In fact there is lot of evidence that, in various parts of the three volumes of capital, Marx himself made that same argument.¹³ Once we take the positive relationship between the rate of exploitation and the organic composition into account, we can no longer agree with Marx's assertion about the law of the tendential fall in the rate of profit. This is because the rate of exploitation might rise to nullify the effect of the rise in the organic composition on the rate of profit. But we can still make a weaker claim: if the elasticity of the rate of exploitation with respect to the organic composition of capital is not too high, then an increase in the latter will lead to a fall in the rate of profit.

Proposition 1: Let Q denote the organic composition of capital. If the elasticity of the rate of exploitation with respect to the organic composition of capital is less than $Q/(1 + Q)$, then any increase in the organic composition will lead to a fall in the rate of profit.

While a mathematical proof of this result can be found in the appendix, the intuition is relatively easy to grasp.¹⁴ The elasticity of the rate of exploitation with respect to the organic composition is the percentage change in the former that is caused by a one percentage change in the latter. It measures the responsiveness of the rate of exploitation to changes in the organic composition of capital. Recall that the rate of profit is the ratio of the rate of exploitation and the organic composition of capital. Thus, if the rate of exploitation is not "too responsive" to the organic composition of capital then an increase in the latter does not cause "too large" a change in the former. Hence, in this case, the effect of the increase in the organic composition will not be nullified by the effect of the increase in the rate of exploitation, so that the rate of profit will fall. Proposition 1 gives the precise conditions under which this intuitive reasoning will be valid. It is worth noting that for large values of Q , the ratio $Q/(1 + Q)$ comes arbitrarily close to unity. Hence, for economies with large values of the organic composition of

¹² After explaining the operation of the law of the tendential fall in the rate of profit, Marx went on to outline many counteracting tendencies which might thwart the operation of the law so that we might not observe the rate of profit falling. The discussion here relates to the validity of the logic behind the law itself. Hence, the existence of counteracting tendencies is not relevant for the argument.

¹³ For examples of quotations, see Sweezy (1942, pp. 101-102).

¹⁴ A variant of this result was highlighted by Sweezy (1942) in the footnote on page 102.

capital, Q , if the rate of exploitation has an elasticity with respect to Q that is lower than unity, i.e. the rate of exploitation is inelastic, then exogenous increases in the organic composition will lead to a fall in the rate of profit.¹⁵

3.3.2. Value or Physical Quantities

The second question is more difficult to answer: why must the organic composition of capital rise with capital accumulation? Marx's argument, developed in Chapter 25 of Volume One of Capital and in Chapter 13 in Volume Three of Capital, and elaborated by later scholars like Mattick (1974) and Shaikh (1978), starts from the recognition that capital accumulation is mediated and enforced in capitalist economies through the competitive struggle between capitalist firms. The process of competition between capitalist firms creates strong incentives for finding and adopting cost-reducing methods of production. After all, reducing the cost of production is, for capitalist firms, the surest way of keeping ahead in the competitive struggle. This is because reduction in the cost of production can increase profits and expand market shares. Since cost of labour power is an important component of the total cost of production, the search for cost reduction often ends up in the adoption of labour saving technical change, i.e. mechanization, or the replacement of workers with machines. Thus, with the progress of capitalist production, the ratio of machines to workers rises. This is why, these authors would argue, the process of capital accumulation is accompanied by a rise in the organic composition of capital.

While it is true that capital accumulation leads, on the whole, to an increasing mechanization of the production process, this does not imply that the organic composition of capital rises. The increasing mechanization of the production process is manifested as each worker working with more machines and converting more raw materials into finished products.

[T]he same number of workers or the same quantity of labour-power that is made available by a variable capital of a given value, as a result of the specific methods of production that develop within capitalist production, sets in motion, works up, and productively consumes, within the same period, an ever growing mass of means of labour, machinery and fixed capital of all kinds, and raw and ancillary materials... (Marx, 1991, pp. 318)

Thus, in purely physical terms, the ratio of non-labour and labour inputs in production *might* rise because “an ever growing mass of means of labour” is used by each worker. But this does not imply that

¹⁵ A strand in the literature has tried to bypass the problem captured by Proposition 1 with two strategies: (1) by using the ratio of constant capital to value added, $C/(V + S)$, instead of the organic composition of capital, C/V (Wright, 1977); and (2) by using asymptotic arguments (Shaikh, 1978). None of these strategies offer valid responses to the problem, and the essential difficulty remains unaddressed. A better strategy, in my opinion, is to acknowledge the problem and offer the weaker version of the claim of the law of the tendential fall in the rate of profit that is captured in Proposition 1. Of course a deeper problem might be the fact that both the organic composition of capital and the rate of exploitation are endogenous to the process of capital accumulation. Thus, arguments which rely on movements of either of these two components of the rate of profit might need to be supplemented by deeper arguments about truly exogenous factors.

the organic composition of capital also rises. This is because the organic composition of capital is a value magnitude. It is the ratio of the value of constant capital and the value of variable capital. The fact that the physical ratio of non-labour and labour inputs in production might rise does not imply that the corresponding value ratio will also rise. There are two complications to consider.

First, the process of technological change reduces the value of all commodities, including all types of means of labour, i.e. machinery and other fixed capital. Thus, even when each worker works with a larger *mass* of means of labour, the fall in the value of each unit of the means of labour might very well imply a fall, rather than a rise, in the organic composition of capital.¹⁶ Second, one of the implicit assumptions hidden in the argument is that the only way to reduce the cost of production is by the replacement of labour with machines. But that is not necessarily true. The search for cost reduction by capitalist firms might also lead to the replacement of existing machines with better quality machines or in a more economical use of the existing means of labour. Both these arguments rule out the conclusion that the organic composition of capital *must necessarily rise* with capital accumulation.

There is an additional, and probably deeper, issue to consider. If capital accumulation and the accompanying technological progress leads to a fall in the rate of profit, then why would profit-maximizing capitalist firms undertake technological innovations and accumulate capital in the first place? While this issue has drawn enormous scholarly attention since the justly famous contribution of Okishio (1961), the essential issue had been raised earlier by Ladislaus von Bortkiewicz in 1907 and Kei Shibata in the 1930s.¹⁷

3.3.3. The so-called Okishio Theorem

Capitalist firms, argued Okishio (1961), are not concerned about labour productivity growth or the social impacts of technological change. The primary motivation for the adoption of new techniques of production by capitalist firms is to reduce the cost of production – so as to get, and keep, ahead in the competitive struggle. This implies that capitalist firms will only adopt new techniques of production – during the process of capital accumulation and under the pressure of competition – that reduces the cost of production. Reduction in the cost of production implies an increase in the rate of profit. Hence, argued Okishio (1961), the process of technical change and capital accumulation will increase the rate of

¹⁶ There is the additional problem, which I am ignoring for the moment, of making sense of a quantity like the “mass of means of labour” used by a worker. While such a quantity is easy to define and measure in a model of a one-commodity economy, it becomes complicated in a multi-commodity world. How would one compare the “mass of means of labour” used by a worker in cotton production versus in car production, when the machines and raw materials might be very different physical commodities? How would one track such a ratio over time if the physical commodities used as raw materials and means of labour changes? There is no way to define such a ratio other than to use some normalization, either using labour-values or monetary prices. That is why the concept of the “technical composition of capital”, which is supposed to capture the ratio of the “mass of means of labour” and number of workers is theoretically vacuous.

¹⁷ The exact reference for Ladislaus von Bortkiewicz’s work can be found in Sweezy (1942, pp. 104), and the reference for Kei Shibata’s work can be found in Okishio (1961, pp. 85).

profit, not reduce it, as Marx had claimed. This means that the law of the tendential fall in the rate of profit that Marx outlined in Volume Three of *Capital* is not valid.

Once we engage with the details of Okishio's (1961) argument, we discover two problems. First, he seems to implicitly suggest that profit-maximizing behavior of capitalist firms is incompatible with a decline in the average rate of profit. In making this claim, he ignores the difference between the rate of profit earned by an individual capitalist firm and the average rate of profit for the whole economy (or a sector of the economy). Second, his conclusion about the rise in the rate of profit rests on a specific assumption about the movement of the real wage rate. Once we remove that assumption, his main conclusion will be significantly modified.

Let us start with the first problem: it is possible for the adoption of a new technique of production to increase the rate of profit earned by the individual capitalist firm that is the innovator (the first firm to adopt the new technique) and yet to lead to a fall in the average rate of profit. In fact that is precisely how Marx had described the dynamics of technical change under capitalist relations of production in Chapter 12, Volume One of *Capital* (Marx, 1990, pp. 433-437). The innovator firm, by searching for and adopting a new technique of production, is able to produce the relevant commodity at a cost that is lower than the social average. Since the value of the commodity is determined by the social conditions of production, it remains unchanged when one capitalist firm adopts a new technique of production. Hence, by selling the commodity at its ruling price (which is its value in the context of Volume One of *Capital*), the innovator firm is able to earn a super-normal rate of profit. The super-normal profit attracts the attention of other capitalists, who try to imitate the innovator. Gradually there is diffusion of the new technology across the capitalist economy, and the new technique of production is adopted by other firms. When the new technique becomes widely used it defines the new social condition of production and determines the value of the commodity and the average of profit. Under certain conditions it is possible for the *average* rate of profit to be lower after the new technique is adopted by all capitalist firms than what prevailed before its adoption. Considering the conditions under which this might happen takes us to the second problem in Okishio's (1961) argument.

Okishio (1961) makes an important assumption in his analysis: the real wage rate remains unchanged even as the economy moves from using the old to using the new technique of production. This is the key assumption that drives his result about the rise in the rate of profit, and it is easy to see why. Since the new technique of production reduces the overall cost of production, it implies a fall in the value of the commodity.¹⁸ Hence, a larger quantity of use-values can be produced with the same amount of labour or the same amount of use-values can be produced with the same amount of labour. If the real wage rate remains unchanged, the total increase in the productivity of labour would be appropriated by capitalists. Hence, it is not surprising that, in such a case, the rate of profit would rise, as Okishio (1961) argues. But there is no reason why the real wage rate must remain constant during the process of technical change. The real wage rate is the result of a complex bargaining process between capitalists and workers and there are no economic arguments that can rule out the possibility of an increase in the

¹⁸ Recall that the value of a commodity is the total labour directly and indirectly required to produce it, under the currently prevailing social and technological conditions of production.

real wage rate during the process of technical change. If conditions in the labour market are such that workers are able to push for an increase in the real wage rate, Okishio's (1961) result might no longer emerge, and the rate of profit might even fall.

One way to summarize the discussion on the issue of technical change and the rate of profit is to note that claims made by Marx (1990, 1991) and Okishio (1961) are special cases that can be accommodated in a more general framework. The effect of technical change on the rate of profit cannot be determined *a priori*, so that neither Marx's claim about the falling rate of profit nor Okishio's claim about the rising rate of profit is unconditionally true. Rather, the effect of technical change on the rate of profit depends on what happens to the real wage rate in comparison to changes in the productivity of the labour and the non-labour inputs to production entailed by the process of technical change. In fact we can be even more specific. In a simple model of the economy that produces only one commodity – Ricardo's corn model or models used in modern growth theory, both orthodox and heterodox – we can identify a threshold value of the growth rate of the real wage rate – let us call it the Marx-Okishio threshold – that allows us to identify conditions for the validity of Marx and Okishio's contrasting claims.

Proposition 2. Let β_1 and β_2 denote the rates of growth of labour productivity and capital productivity (the output-capital ratio), respectively, which is associated with a new technique of production; and let γ denote the organic composition of capital of the new technique of production evaluated at prices that prevailed prior to the economy-wide adoption of the new technique of production. In such a setting, we can define a Marx-Okishio threshold as follows: $\alpha^ = \beta_1 + \gamma\beta_2$. Using the threshold, we have the following:*

1. *Marx's Result: If the actual growth rate of the real wage rate is higher than α^* , then the average rate of profit falls after the adoption of the new technique of production;*
2. *Okishio's Result: If the actual growth rate of the real wage rate is lower than α^* , then the average rate of profit rises after the adoption of the new technique of production.*

A proof of Proposition 2 can be found in the appendix, but here let us dwell on the underlying intuition. If the rate of growth of the real wage is relatively high, i.e. above the threshold value identified in Proposition 2, then a relatively large share of the fruits of productivity growth coming from technological change is captured by the working class. In such a scenario, the rate of profit falls and we are in a world where Marx's claim holds true. On the other hand, if the growth rate of the real wage rate is low, then most of the benefits of productivity growth is captured by the capitalist class. This is manifested in the rise of the rate of profit, and we are in a world where Okishio's claim holds true. There are no theoretical or empirical grounds to believe that capitalist economies function in a way that makes either Marx's results or Okishio's results always true. Depending on the exact mechanisms that determine the growth of the real wage rate, capitalist economies might function in either fashion.

Before concluding the discussion of the Okishio theorem it is also worth noting the structural similarity between Marx's argument about the law of the tendential fall in the rate of profit and contemporary discussion of coordination problems in capitalist economies. The pressure of competition puts each capitalist in a prisoner's dilemma-like situation. If she forgoes the chance to adopt a cost-cutting

technique of production, she will lose in the competitive struggle. This ever present threat of loss, which in extreme cases can mean bankruptcy, will ensure that capitalists will always search for and adopt cost-cutting techniques of production. It will also mean that despite the best efforts of the innovator capitalist, the new technique of production will diffuse through the economy. But neither the innovator capitalist nor the imitator capitalists can internalize the impact of technical change on the labour market into their decision making process. In some cases, it is possible for the impact of technical change on the real wages (or the value of labour power) to be such that the average rate of profit falls, and all capitalists lose (Foley, 1986, pp. 131).

4. Crisis of Excess Surplus Value

In a crisis of deficient surplus, the rupture in the economy-wide circuit of capital is proximately caused by a fall in the rate of profit, which is, in turn, caused by an insufficiency of aggregate demand. There is a long tradition within Marxian political economy, which I will call the underconsumptionist tradition, which has kept the problems of aggregate demand at the very center of analysis, and it is the primary purpose of this section to discuss the arguments in this tradition. While problems of aggregate demand become salient in discussions of crises of excess surplus value, there have also been some attempts to follow Marx's discussion on finance in Volume Three of *Capital* in developing a parallel argument regarding a mechanism that becomes important in a crisis of excess surplus value: financial fragility (Duménil and Lévy, 2013). The analysis of problems of financial fragility in capitalist economies from within a Marxian framework is in its infancy. Hence, while I offer some brief remarks on the problems of financial fragility, my main comments on the crisis of excess surplus value will be reserved for a discussion of the tendency towards underconsumption.

4.1. Underconsumptionism

The key claim of the underconsumptionist strand of thinking, both within and outside Marxian political economy, is that, left to itself, the capitalist system is unable to generate enough aggregate demand to sustain a positive rate of growth of the system. The underconsumptionist argument comes in two variants, with two versions of the first variant. The first variant works with a two department conception of the economy, with one department producing means of production and the other producing means of consumption. While we find a rather crude version of this first variant in the writings of early underconsumptionists like Sismondi and Rodbertus, a more sophisticated version was developed in Sweezy (1942). The second variant moves away from the two department conception of the economy and instead presents its argument within an aggregate, macroeconomic framework more in line with the work of Keynes and Kalecki than the framework used by Marx. A well-developed version of this variant of the underconsumptionist argument is presented in Baran and Sweezy (1966). I discuss each of these strands in detail below.

4.1.1. First Variant of Underconsumptionism

There are at least two versions of the first variant of the underconsumptionist argument, and in presenting them I draw on Shaikh (1978).

First Version

The first version, developed by early underconsumptionists, advances the claim that a typical capitalist economy is incapable of self-expansion on the basis of internally generated aggregate demand. There are two key ingredients of this argument, the demand gap and vertical integration, and let us take these up in turn.

To understand the notion of “demand gap”, let us note that, starting from the total (gross) output of commodities, if we remove the portion that is used up in the production process, we will get the net output. The net output corresponds, and is equal in value, to the total income earned by all agents involved in the production process. If, for simplicity, we assume that there are only two classes involved in production, capitalists and workers, then the sum of all profit income (surplus value) and all wage income (variable capital) is equal to the value of the net output.

To bring the focus on the problem of demand, let us ask the following question: who will purchase the net output? A portion of the net output will be purchased by workers – using their wage income. Even if workers spend all their income on consumption, they will not be able to purchase the total net output. This is because the value of net output is the sum of wages and profits, and since the latter is some positive quantity in capitalism, the value of net output is larger than the total wages. This leaves a “demand gap”, i.e. a portion of the net output that remains to be sold even after all wages have been spent.

To grasp the concept of vertical integration, let us divide total social production into two departments. The first department, called Department I, consists of capitalist firms that produce means of production, including machines, raw materials, fuel, etc. The second department, called Department II, produces means of consumption, like food, clothing, health care, entertainment, etc. Early underconsumptionist thinkers conceived of Department I as being completely subsumed under Department II. Thus, according to this line of thinking, the output of Department I is only used as “inputs” to the production process in Department II. An alternative way of stating this is to say that the economy is vertically integrated.

There is an important implication of this conception of a vertically integrated capitalist economy. Since the output of Department I functions only as inputs to the production process in Department II, this means that the net output is only composed of the output of Department II. Let us call the part of net output that remains after all the wage income has been spent as the “surplus product”. Since the surplus product is a part of the net output, this also implies that the surplus product is composed only of the output of Department II.

We are now ready to lay out the basic argument of the first variant of underconsumptionism. Consider a capitalist economy with a zero demand gap. The fact that the demand gap is zero means that the total surplus value is used to purchase the surplus product – that is the only way that the demand gap can be closed. Since the surplus product is composed entirely of the output of Department II, it means that the

whole surplus value has been spent on purchasing consumer goods. Hence, there is zero investment in this economy, i.e. none of the surplus value was used to purchase means of production and increase the scale of production. This means that the economy has no capital accumulation and, hence, can only have zero growth. A capitalist economy, according to this early underconsumptionist vision, is not capable of sustaining positive rates of growth on the basis of internally generated aggregate demand.

We can arrive at the same conclusion by a slightly different route. Let us consider a capitalist economy with a zero demand gap. As we have noted above, this means that all the surplus value is used by capitalists to purchase consumer goods. What will happen if some capitalists save and invest part of the surplus value to increase the scale of production? If capitalists save a part of the surplus value and invest it, i.e. use a part of the surplus value to purchase means of production and labour power and increase the scale of production, this will have two effects. First, there will be a net decline in the demand for the output of Department II. Why? There will be a decrease in demand due to the saving out of surplus value; but there will be an increase in demand given by the part of this saving that is used to purchase labour power (which will be used by workers to purchase consumption goods). Since the former is larger than the latter (as long as some of the surplus value is used to purchase additional means of production), there will be a net decline in the demand for the output of Department II. The second effect of saving/investment of surplus value will be an increase in the output of Department II (unless all the additional investment goes towards the production of additional means of production, an unlikely scenario). Thus, the output of Department II increases exactly at the same time as demand declines, leading to a problem of realization. The overall result will be an increase in excess capacity in Department II, leading to a reduction in investment and growth. Thus, the impulse towards capital accumulation and growth, coming from saving and investment, is self-defeating. The natural state of a capitalist economy, the early underconsumptionists argued, is a state of stagnation.

A particularly interesting extension of this argument comes from considering the problem of growth of capitalist economies when there is an increase in income inequality. This could come about, for instance, due to stagnant real wages in the face of growing labour productivity. If labour productivity increases but real wages do not, that increases the share of surplus value in total value added. Since the value of surplus product is equal to the surplus value, a shift in the distribution of income away from wages will increase the surplus product. By the logic of the underconsumptionist argument, this would widen the demand gap and heighten the tendency towards stagnation in the capitalist economy.

How do we assess the strength of this version of the underconsumptionist argument? The basic problem in this variant of the underconsumptionist argument is the faulty conception of the surplus product – understood to be composed entirely of consumer goods – that comes from conceiving the economy as a vertically integrated production system. If the economy is conceived as a vertically integrated system, then the output of Department I (machines, for instance) can only be used as inputs into Department II. But this is not true. Machines can be used for producing machines, i.e. the output of Department I can be used as inputs into Department I itself. Thus, conceiving the economy as a vertically integrated production system is problematic.

A better way to conceptualize the relationship between Department I and II is to use the reproduction schemas that Marx developed in Chapters 20 and 21 of Volume Two of *Capital* (Marx, 1992). If we use Marx's reproduction schemas, we can demonstrate that a capitalist economy is able to generate, under certain conditions that relate to the *proportionality between the two departments*, adequate aggregate demand to purchase the total output of commodities. We can demonstrate this result not only for a capitalist economy undergoing simple reproduction, i.e. a zero growth system, but also for a capitalist economy undergoing reproduction on an expanded scale, i.e. when it grows at some positive rate of growth. In fact, savings and investment out of surplus value, the main source of problem for the capitalist economy as depicted in the underconsumptionist argument, now becomes the driver of growth of the system. Not only is the system capable of generating self-sustaining positive rates of growth, if the ratio of the sizes of the two departments is "correct", then the system can smoothly reproduce over time, i.e. it would be plagued neither by problems of shortage nor of glut on its growth path.

To state this result formally, we need to specify three parameters characterizing the economy, related, respectively, to the technology of production, the degree of exploitation of workers, and the investment propensity of capitalists.¹⁹ In writing these parameters, we will use subscripts *I* and *II* to identify the two departments.

Let k_I and k_{II} denote the share of variable capital in total capital outlays in the two departments; these parameters, being functions of the organic composition of capital, characterize the technology of production used in the two departments. A higher value of the organic composition of capital, which could come about due to an increased capital intensity of production, leads to a lower share of variable capital in total capital outlays. Let e denote the rate of exploitation (ratio of surplus value and variable capital) that is common in both departments; a higher value of the rate of exploitation implies that more surplus value is extracted from workers for each unit of variable capital advanced. Let p_I and p_{II} refer to the share of surplus value reinvested into production in the two departments; these characterize the investment behavior of capitalists in the two departments, with higher values of p_I or p_{II} capturing a higher propensity to invest.

Proposition 3: Let V_I and V_{II} refer to the variable capital in departments I and II, respectively. If $k_{II}p_{II} = k_Ip_I = \eta$ and the ratio of the sizes of the two departments is given by

$$\frac{V_{II}}{V_I} = \frac{1+e-ep_I(1-k_I)}{\left[\frac{1-k_{II}}{k_{II}}\right]+ep_{II}(1-k_{II})} \quad (4)$$

then the capitalist economy can smoothly reproduce on an expanded scale at the rate of growth $g = \eta e$.

A proof of Proposition 3 is available in the appendix. Here, let us consider the underlying intuition. Given the parameters that capture technology, exploitation and investment behavior in the two departments,

¹⁹ The formal analysis in this section draws on Harris (1977, chapter 10) and Foley (1986, chapter 5).

the right hand side of the expression in (4) is some positive number.²⁰ Thus, proposition 3 shows that if the sizes of the variable capitals in the two departments maintain the proportion given by the right hand side of (A3), then the capitalist economy can ensure smooth reproduction on an expanded scale. How do we know this? The condition comes from an algebraic manipulation of the condition for equilibrium – equality between demand and supply – in the market for the output of Department I (for details see section A3 in the Appendix). Hence, if this condition is satisfied, it will imply that the total demand for means of production will be exactly equal to its total supply, both expressed in terms of value. Moreover, since total social production is broken up into two departments, equilibrium in the market for means of production (the output of Department I) will automatically imply equilibrium in the market for means of consumption (the output of Department II).²¹ Thus, there will be neither excess demand nor excess supply in the market for means of production and in the market for means of consumption, which means that there will be smooth reproduction of the whole system of production.

The fact that the condition about the proportional sizes of the two departments given in (4) ensures smooth reproduction also highlights a possible mechanism for the emergence of crisis: *disproportionality between the two branches of production*. If the two branches are not of sizes that satisfy the condition given in Proposition 3, then smooth reproduction of the system will not be possible. Either there will be excess supply for means of production, coupled with excess demand for consumer goods, or the economy will face an excess demand of means of production as also an excess supply of consumer goods. In either case, there will be an overproduction of commodities, and the economy will be plunged into a crisis – a crisis of disproportionality, as highlighted by Tugan-Baranowsky (see Sweezy, chapter X and XII).²²

Second Version

The second, and more sophisticated, version of the first variant of the underconsumption argument is found in Sweezy (1942). The setting of his argument is the same as the one we have been using: a two department capitalist economy undergoing expanded reproduction. Following classical economists, Sweezy (1942) assumes that all wages are consumed. On the other hand, surplus value is used for four purposes: a part is consumed to keep capitalist consumption at the same level as before, another part is used to increase capitalist consumption, a third part is used to augment constant capital and a fourth part to increase variable capital. The main argument in Sweezy (1942) is developed in two steps.

The first step relates to the implications of capitalist relations of production on the growth of demand for consumption goods. It is undoubtedly true that capitalists want to increase profits to the maximum possible extent. This goal is usually achieved through two mechanisms: (a) through an increase in

²⁰ Using the fact that k_I , k_{II} , p_I and p_{II} are fractions and lie between 0 and 1, it is easy to show that both the numerator and the denominator are positive.

²¹ In economics, this is known as Walras' Law.

²² An alternative formalization of the circuit of capital model can be found in Foley (1982). In that model, increases in production and realization lags can also open up a demand gap. The demand gap can either slow down the steady state growth rate of the economy or lead to a crisis.

accumulation as a share of surplus value, and (b) by an increase in investment as a share of accumulation. The latter comes about as a direct result of the increasing mechanization of the production process, itself an outcome of the competitive struggle between capitalists.

Consumption demand comes from two sources, wages (all of which is consumed) and the share of surplus value that is not accumulated. As capitalism develops, the rate of surplus value rises, which implies that wages decline as a share of value added. On the other hand, the competitive pressure ensures that capitalists accumulate a larger share of surplus value. Hence, the share of surplus value that is not accumulated, and which is the source of capitalist consumption demand, declines over time. Thus, the ratio of the growth rate of consumption demand to the growth rate of the capital stock declines over time.²³

The second step of the argument looks at the logic of production from a technological perspective. Sweezy (1942) believes that there is a constant and stable ratio between the capital stock and the potential output (or supply) of consumption goods. This is the key underlying assumption about technology and implies that the ratio of the growth rate of the supply of consumption goods to the growth rate of the capital stock is a constant. If we bring the results from the two steps together, we see the key point of the argument: the ratio of the growth rate of the demand for consumption goods to the growth rate of the supply of consumption goods falls over time. Hence, at some point, the supply of consumption goods overshoots demand, pushing the economy towards an acute realization problem.

How do we assess Sweezy's (1942) argument about the tendency towards underconsumption? There is a basic flaw in the above argument and relates to a faulty assumption about technology (Shaikh, 1978). There is no reason why the capital stock must maintain a constant and stable relationship to the potential supply of consumption goods. The stock of capital can be increased, with investment, to produce more means of production, in which case the potential supply of consumption goods would not change. In such a case, the ratio of the capital stock to the potential supply of consumption goods would fall, contradicting the second step of Sweezy's (1942). The upshot is that Sweezy' (1942) theory of underconsumption rests on a questionable theory of production and technology. Once that assumption is abandoned, the theory's main conclusion about the tendency towards stagnation in capitalist economies falls apart.

4.1.2. Second Variant of Underconsumptionism

The second variant of the underconsumptionist argument works in an aggregate, Keynes-inspired macroeconomic framework, eschewing the distinction between Department I and Department II. A cogent version of this argument is found in the influential work by Baran and Sweezy (1966). The key claim of their work is a forceful argument that monopoly capitalism is characterized by a strong tendency towards stagnation. What is the reason for this?

²³ Note that the growth rate of the capital stock is called investment.

“[Monopoly capitalism] tends to generate ever more surplus, yet it fails to provide the consumption and investment outlets required for the absorption of the rising surplus and hence for the smooth working of the system. Since surplus which cannot be absorbed will not be produced, it follows that the *normal* state of the monopoly capitalist economy is stagnation.” (Baran and Sweezy, 1966, pp. 108; emphasis in original).

The economic reasoning underlying Baran and Sweezy’s (1966) claim can be understood with the help of a diagram that is adapted from their analysis and depicted in Figure 1. The horizontal axis in Figure 1 measures the economy’s capacity utilization rate, which is defined as the ratio of actual output to capacity output. On the vertical axis of Figure 1, we measure two variables: the economic surplus, and the total expenditure to absorb the surplus. The economic surplus is defined as the difference between the total output and the necessary costs of producing that output, the latter including both labour and non-labour costs of production. The total expenditure to absorb the surplus is defined as the level of expenditure in the economy that is over and above the expenditure out of wage income. Assuming, along classical political economy lines that all wages are consumed, the expenditure out of wages is equal to the full value of the wage income. Thus, the expenditure to absorb the surplus becomes the total expenditure over and above the total wage income. This can, in turn, be seen to be composed of three categories of expenditures: (a) consumption expenditure of capitalists; (b) investment expenditure by capitalists; and (c) wasteful or unproductive expenditure, i.e. expenditure to support unproductive activities, where the latter refer to all activities that do not generate surplus value.

In Figure 1, economic surplus is represented by the upward sloping lines S_1, S_2 and S_3 , which we call the *surplus curves* and where the subscript refers to consecutive time periods. At any point in time, for instance in time period 1, the economy has a given capacity base for production. For this given production capacity, the economic surplus is an increasing function of the capacity utilization rate and is represented in Figure 1 by the upward sloping lines S_1 . Economic surplus depends positively on the capacity utilization rate because of the presence of overhead costs – cost which do not vary with the level of output. Thus, when the level of output increases, the overhead cost per unit of output falls. If the variable cost remains more or less constant, then for any price level, the surplus increases with the level of output. For a given capacity base, the capacity utilization rate increases with the level of output, so that the economic surplus rises with the former. This is what imparts the upward slope to the line S_1 . Over time, the economic surplus curve shifts up, going from S_1 to S_2 to S_3 , and so on, and this long run movement is the key element of Baran and Sweezy’s (1966) argument. But before we discuss the long run dynamics, let us look at the short run equilibrium.

At any point in time, the level of expenditure to absorb the surplus is an upward function of the capacity utilization rate and is represented in Figure 1 as the upward sloping *expenditure curve*, E . One of the components of the expenditure stream that absorbs the surplus is investment expenditure by capitalist firms, and this is likely to be an increasing function of the capacity utilization rate. As capitalist firms operate with a higher proportion of their full capacity, they are going to be more likely to start planning for adding to that capacity, and hence investing. Thus, even if the other two components of expenditure do not depend on the capacity utilization rate, the total expenditure to absorb the economic surplus

would be an increasing function of the capacity utilization rate. This is what gives the upward sloping expenditure curve E in Figure 1.

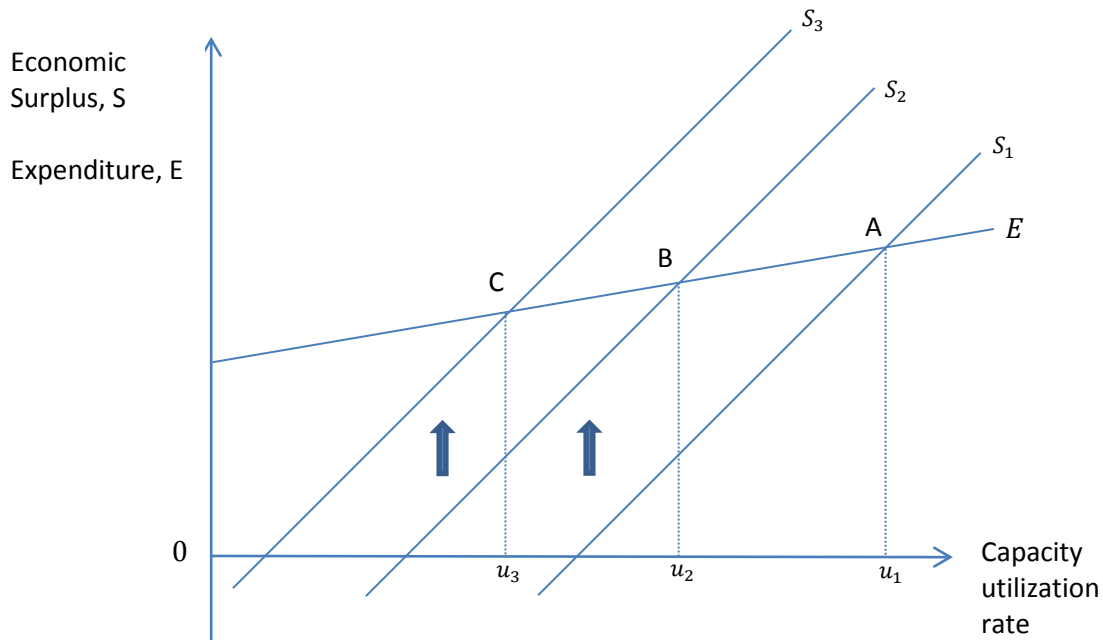


Figure 1: Short run and long run equilibrium in a monopoly capitalist economy. The upward sloping surplus curves, S_1 , S_2 and S_3 , represent the amount of economic surplus generated by the system as a function of the capacity utilization rate. The upward sloping expenditure curve E represents the level of expenditure that can absorb the economic surplus as a function of the capacity utilization rate. The intersection of the two curves gives the equilibrium rate of capacity utilization in the economy in the short run. Over time, the surplus curve shifts up. This leads to a downward drift of the equilibrium level of capacity utilization.

The short run equilibrium of the economy is given by the intersection of the surplus curve and the expenditure curve. For instance, in period 1, the intersection is given by the point A , so that the corresponding level of the capacity utilization rate is represented by u_1 . The meaning of the short run equilibrium, given by the capacity utilization rate u_1 , is the following: the structure of expenditures in the economy is such that it can only support a utilization rate of u_1 because only at this rate of utilization of production capacity does the economy generate enough demand (expenditures) to absorb all the economic surplus. In the long run, the surplus curve shifts up, moving from S_1 to S_2 and then to S_3 and so on. The equilibrium in the economy moves from A to B to C and so on. The corresponding level

of the equilibrium capacity utilization rate falls from u_1 to u_2 to u_3 and a monopoly capitalist economy sinks “deeper and deeper into a bog of chronic depression.” (Baran and Sweezy, 1966, pp. 108).²⁴

It is obvious that the key mechanism pushing the economy towards stagnation, i.e. towards lower and lower rates of utilization of production capacity, is the constant upward movement of the surplus curve. What is the reason for that? Monopoly capitalism is a system made up of giant corporations. In sharp contrast to firms in competitive capitalism, which were price takers, the giant corporations making up monopoly capitalism are price makers, i.e. they “can and do choose what prices to charge for their products”. In a system composed of price takers, “price competition is banned as a legitimate weapon of economic warfare”. Prices of products are set so as to maximize the profits of the whole group of corporations producing that product or its close substitutes. While this can take various forms like cartelization or price leadership, the important implication is that such price setting behavior imparts a pronounced upward bias “into the general price level in a monopoly capitalist economy”. While price competition is banned, that does not mean the end of competition, “it takes new forms and rages on with ever increasing intensity”. The key impact of non-price competition on the system is that it creates and maintains strong pressures on each giant corporation to reduce its costs of production – that is how it can keep ahead of its competitors. Thus, at the aggregate level, prices are downward sticky at the same time as costs of production fall. Therefore, the difference between the two, which is the economic surplus, tends to become larger over time. In Figure 1, this is represented as the upward movement of the surplus curves over time, from S_1 to S_2 to S_3 and so on.

There are several noteworthy points about the above argument.

First, the evolution of the equilibrium capacity utilization rate for the economy depends on the movement of both the surplus curve and the expenditure curve. Even if the surplus curve shifts up over time, the economy might still be able to sustain a high or rising rate of utilization of productive capacity if the expenditure curve also shifts up.²⁵ Hence, without a well worked theory of the evolution of the expenditure side of the economy, the overall argument is incomplete. While a large part of Baran and Sweezy (1966) is devoted to explaining why factors that could absorb the rising surplus is not strong under monopoly capitalism – see chapters 4 through 7 of the book – that part of the argument is not as tightly developed as the part that demonstrates the tendency for the surplus to rise. The arguments about the insufficiency of aggregate demand are developed only informally and the quantitative dimension remains underdeveloped. This is problematic because the key argument is a quantitative one, viz., that expenditure to absorb the surplus (aggregate demand) is *less* than the total amount of economic surplus.

²⁴ In later work, Sweezy (1981) has called this argument a crisis of overaccumulation or a tendency for overaccumulation. The essence of the argument remains the same as found in Baran and Sweezy (1966).

²⁵ An additional problem relates to the slopes of the S curve and the E curve. If the E curve is steeper than the S curve, then an upward movement of the S curve will lead to an increase in the capacity utilization rate. But this problem can be ruled out by resorting to something like a Keynesian stability condition, which would require that the E curve be flatter than the S curve.

Second, the argument about the tendency of the economic surplus to rise over time has an important lacuna. One of the key components of cost is the real wage rate and it is not clear why the real wage rate will have a downward trend in a monopoly capitalist economy. If we visualize the real wage rate as being determined in the labour market through a process of bargaining between firms and workers, then many factors other than the nature of firms – competitive versus oligopolistic – would impact the outcome. The organization of workers in unions, legal provisions relating to hiring and firing of workers by firms, unemployment insurance and other social security benefits, and other such factors would impact the level of real wages. Hence, despite the existence of oligopolistic firms, the real wage rate might not fall over time. If that is the case, then the economic surplus would not necessarily rise over time. The fact that real wage rates in most advanced capitalist economies rose in a robust manner for close to two decades after the second world, but then stagnated for the next three decades, suggests that there is no necessary link between monopoly capitalism and the evolution of the real wage rate. Since the real wage is a significant component of the cost of production, this raises a serious question about the key claim of the above argument that there is a tendency for the economic surplus to rise over time in monopoly capitalist systems.

A corollary of the first point is also worth highlighting. The main thrust, and the key analytical content, of the above argument is about the supply side of the economy. It relates to why the juxtaposition of oligopolistic pricing behavior and non-price competition between giant corporations leads to the generation of increasing amounts of surplus. Thus, unlike many interpretations which understand the above argument as a demand-side argument (for instance, Foster 2016), it seems that one should understand it as primarily as a supply-side argument. Re-stated in the language of contemporary macroeconomics, the above argument claims that the aggregate supply curve shifts upwards over time, and absent an equally large upward movement of the aggregate demand curve, the economy's equilibrium output falls over time. It is as if the monopoly capitalist economy is being hit by an unending stream of negative supply shocks – like the oil price rises of the 1970s or the 2000s – with stagnant aggregate demand. At best, that is a rather odd way to understand the evolution of a capitalist, competitive or monopolistic, macroeconomy.

4.2. Problems of Financial Fragility

In a situation of excess surplus value, the financial system comes under severe strain (Foley, 2012). This is because the ability of the financial system to channelize surplus value into expenditure streams for the purchase of produced commodities, i.e. newly produced goods and services, gets eroded. One reason for this outcome is that in a situation of excess surplus value, the financial system is flush with funds, so that interest rates are low. When interest rates are low in a capitalist economy, it encourages speculative activity with borrowed funds, diverting funds from away from the expenditure on the produced output of commodities. Instead, the speculative activity is usually directed at purchasing key assets, like land, or shares of stock, in the hope of making capital gains.

For some time, which can even range over many years, the speculative activity becomes self-fulfilling. Economic agents use borrowed funds to purchase assets with the aim of making capital gains. If this

activity is undertaken by a large fraction of the relevant set of economic agents, the demand for the asset rises and leads to a rise in its price. Thus, the initial expectations of the speculators are fulfilled, and it draws in more funds into the speculative activity. Thus, the speculative activity both feeds on and supports a bubble in the price of the relevant asset, i.e. a price level that is largely unrelated to the real use of the asset.

There are two effects of the speculation-driven asset price bubble. First, it leads to a 'wealth effect' that boosts debt-financed expenditures. Since the price of the relevant asset increases rapidly, owners of that asset feel wealthy. Hence, they use that asset as collateral to borrow and spend. This boost to spending allows capitalist firms to sell commodities and make profit. In effect, the problem of aggregate demand that plagues a capitalist economy generating excess surplus value is solved temporarily. Second, it increases the financial fragility of the system because it increases the share of debt-financed expenditures in the economy. Speculative activity that aims to make capital gains is largely financed with borrowed funds. Moreover, the debt-financed expenditures on commodities are also, most obviously, rely on borrowed funds. Thus, the overall result is an increase in the share of debt-financed expenditures among all expenditures.

Asset price bubbles inevitably deflate, often triggered by some event in the larger economy. As expectations turn sour, speculators leave the asset in droves, driving down its price precipitously. This brings to the fore a long and complicated chain of payment commitments that cannot be met – because they were based on the expectations of higher, or even increasing, price of the asset. Payment failures cascade through the system, roiling balance sheets and ushering in a full blown financial crisis.²⁶ The crisis in the financial system manifests itself in outright reduction of volumes of credit, increases in the cost of credit, deleveraging by economic agents to repair balance sheets, fire sale of assets, and other related phenomena. All these processes lead to drastic falls in expenditures, reductions in aggregate demand and eventually push the economy into an economic crisis.²⁷

5. Conclusions

Together, the interlinked processes of generation, realization and accumulation of surplus value provide the primary motive force behind the dynamics of a capitalist economy. Smooth reproduction of the

²⁶ "In a system of production where the entire interconnection of the reproduction process rests on credit, a crisis must evidently break out if credit is suddenly withdrawn and only cash payment is accepted, in the form of a violent scramble for means of payment. At first glance, therefore, the entire crisis presents itself as simply a credit and monetary crisis. And in fact all it does involve is simply the convertibility of bills of exchange [i.e. promises to pay] into money. The majority of bills represent actual purchases and sales, the ultimate basis of the entire crisis being the expansion of these beyond the social need. On top of this, however, a tremendous number of these bills represent purely fraudulent deals, which now come to light and explode; as well as unsuccessful speculations conducted with borrowed capital, and finally commodity capitals that are either devalued or unsaleable, or returns that are never going to come in." (Marx, 1991, pp. 621).

²⁷ For discussions of the role of the financial system in the 2008 crisis, see Dumenil and Levy (2011) and Kotz (2015).

system requires that the unity of these three key processes be maintained. And to do so, the system needs to produce the 'correct' amount of surplus value, neither too little nor too much. Since it is an unplanned and contradictory system, resting ultimately on the contradiction between use value and exchange value of the commodity form of production, a capitalist economy is neither able to know the 'correct' amount of surplus value nor produce it any sustained manner. It is forever plagued by one or the other of two problems: production of too much surplus value or production of too little surplus value. Under the former scenario, the economy faces a crisis of excess surplus value, and in the latter case, a crisis of deficient surplus value.

Both types of crisis represent deep and prolonged interruptions of the economy-wide circuit of capital and manifest themselves as crises of overproduction, the accumulation of stocks of unsold and unsalable commodities. The proximate cause of both types of crisis is a drop in the rate of profit, which leads to sharp cutbacks in capital outlays, thereby interrupting the circuit of capital. In a crisis of excess surplus value, the drop in the rate of profit arises due to a lack of aggregate demand so that commodities cannot be sold at prices to realize the full value (and surplus value) of commodities. In a crisis of deficient surplus value, on the other hand, the rate of profit falls not because of insufficient aggregate demand but because of technological or social factors. Either the organic composition of capital rises or the rate of exploitation falls, leading to a fall in the rate of profit even when commodities are sold at their full values (or prices of production).

The synthetic account of Marxist approaches to capitalist crises that has been presented in this paper offers a fresh perspective on several important controversies that have marked this literature.

The controversy between proponents of the 'falling rate of profit' crisis tendency and the 'problems of demand' crisis tendency that has raged on for decades seem, from the perspective of the analysis of this paper, rather unproductive and even unnecessary. Capitalist economies are prone to both types of crises, the first when the system generates too little surplus value and the latter when it generates too much. There is no theoretical reason to believe that capitalist economies will be plagued by only one or only the other. Depending on the configuration of the capitalist economy, it can be witness either type of crisis. If this is true, then the task of Marxist research would not be to try to identify *the* crisis tendency that is valid for all situations, but rather to study particular configurations to see which crisis tendency seems more likely – a concrete analysis of a concrete situation.

A second, and related, controversy has focused on the role of aggregate demand in capitalist economies. The analysis of capitalist dynamics carried out with Marx's schemes of reproduction, both in cases of simple and expanded reproduction, demonstrate that capitalism can indeed generate adequate demand for the commodities produced. Thus, simple minded underconsumptionist arguments, like those presented by Rosa Luxemburg, are logically incorrect. But, while recognizing the logical problems of all variants of the underconsumptionist arguments, it is important to also stress the following point that Kalecki (1971) noted: the fact that capitalism *can* generate adequate aggregate demand does not mean that it always *will*. Capitalist reproduction rests on the unity of two separate phases, the generation of surplus value and the realization of surplus value. Since capitalism is not a planned system, there is no automatic mechanism to ensure that all the surplus value that is generated will also be

realized. Hence, the problem of effective demand is very real in capitalism. But to locate the source of the problem, one needs to move beyond underconsumption theory, both from its simple and sophisticated versions, and develop a theoretically informed and empirically grounded theory of long run capitalist investment.

Modern heterodox macroeconomics, working within a framework that comes out of the work of Keynes and Kalecki, has incorporated explicit investment functions into models of growth and distribution. The ‘investment function’ is precisely an attempt to capture the determinants of capitalist expenditure on capital outlays, and thus fills the lacunae in Marxist underconsumptionist arguments. The general conclusion of this literature is that capitalist economies can be either “wage-led” or “profit-led”; economic theory cannot rule out one or the other. Moreover, when a capitalist economy is profit-led, a shift in income towards workers will *not* lead to an increase of the growth rate of the economy. Thus, the key intuition of underconsumptionist thinking – that a shift in income distribution in favor of workers – will ameliorate problems of aggregate demand and boost growth does not always hold even in models of growth that explicitly allow output to be demand-determined. In addition, a large body of empirical evidence seems to suggest that advanced capitalist economies, especially when international trade is taken into account, are profit-led (Barbosa-Filho and Taylor, 2006; Kiefer and Rada, 2015). Thus, a more nuanced view about the role and constraints of demand is needed than the one that comes out of the underconsumptionist tradition.

The third controversy relates to the so-called Okishio theorem, which has been interpreted as disproving Marx’s claim that the average rate of profit has a tendency to fall in capitalist economies. Much of this controversy also seems, with the benefit of hindsight, needless. There are no theoretical grounds to claim that, due to technological change, the rate of profit will have a tendency to always fall (as Marx claimed) or that it will have a tendency to always rise (as Okishio claimed). A careful analysis shows that the impact of technological change on the rate of profit depends crucially on what happens in the labour market. If the real wage rate rises sharply during the period of technological change, the rate of profit will have a tendency to fall; on the other hand, if the real wage rate does not rise fast enough, the rate of profit might rise. The idea that there is no necessary contradiction between the claims advanced by Marx and Okishio, and that whether the rate of profit falls or rises after the adoption of a new technique of production ultimately depends of how the real wage rate behaves, is, surprisingly, present right in the paper by Okishio (1961) – the origin of the whole controversy.²⁸ The same idea was highlighted by Foley (1986, chapter 8). But the subsequent literature, perhaps bent on polemics, has ignored this basic fact.

I would like to end by pointing out that the study of crisis tendencies in capitalist economies is an extremely important and integral part of Marxist political economy. It separates Marxist approaches from both the neoclassical and Keynesian (or post-Keynesian) approaches to the study of the dynamics

²⁸ “... however large the organic composition of capital may become, the general rate of profit must increase without an exception, only if the newly introduced technique satisfies the cost criterion *and the rate of real wage remains constant*. And we can safely say that every production technique introduced by capitalists reduces the cost of production in terms of prevailing prices and wages. Therefore, we must accept the conclusion that every technical innovation adopted by capitalists in basic industries necessarily increase the general rate of profit *unless the rate of real wages rises sufficiently*.” (Okishio, 1961, pp. 92, emphasis added).

of capitalism. This study is rooted in the deep understanding of Marx and Marxists about the historical limitedness of capitalism as a social form of production. While the Marxist tradition has certainly moved beyond its early twentieth century fascination with implausible theories of capitalist breakdown, it will always keep the study of capitalist crises in the center of its theoretical enterprise. For it is mainly during periods of crisis, with the attendant suffering of the vast majority of the working people, that the contradictions of the system break to the surface and emerges anew the possibility of its positive transcendence.

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Appendix

A1. Elasticity of the Rate of Exploitation

Proposition 1: Let Q denote the organic composition of capital. If the elasticity of the rate of exploitation with respect to the organic composition of capital is less than $Q/(1 + Q)$, then any increase in the organic composition will lead to a fall in the rate of profit.

Proof. Let the elasticity of the rate of exploitation with respect to the organic composition of capital be denoted by η_Q^e so that

$$\eta_Q^e \equiv \frac{Q}{e} \frac{de}{dQ}$$

Since the rate of profit is defined as

$$r = \frac{e}{1 + Q}$$

we see that the derivative of the rate of profit with respect to the organic composition of capital is given by

$$\frac{dr}{dQ} = \frac{1}{(1 + Q)^2} \times \left\{ \frac{de}{dQ} (1 + Q) - e \right\} = \frac{e}{(1 + Q)^2} \times \left\{ \eta_Q^e \left(1 + \frac{1}{Q} \right) - 1 \right\}$$

Since $e > 0$ (because capitalism requires positive surplus value), we can see the following result:

$$\frac{dr}{dQ} < 0$$

if and only if

$$\eta_Q^e < \frac{Q}{1 + Q}$$

which completes the proof.

A2. Marx-Okishio Threshold

Proposition 2. Let β_1 and β_2 denote the rates of growth of labour productivity and capital productivity (the output-capital ratio), respectively, which is associated with a new technique of production; and let γ denote the organic composition of capital of the new technique of production evaluated at prices that prevailed prior to the widespread adoption of the new technique of production. In such a setting, let us define the Marx-Okishio threshold as follows: $\alpha^ = \beta_1 + \gamma\beta_2$. Then we have the following:*

1. *Marx's Result: If the actual growth rate of the real wage rate is higher than α^* , then the average rate of profit falls after the adoption of the new technique of production;*
2. *Okishio's Result: If the actual growth rate of the real wage rate is lower than α^* , then the average rate of profit rises after the adoption of the new technique of production.*

Proof. We consider a one good (corn) economy. Since there is only one good, prices of commodities are proportional to their values (the sum of the direct and indirect labour time required for its production). Without loss of generality, we can assume that the value of money is 1, i.e., every unit of money (say dollar) is equivalent to 1 hour of (socially necessary abstract) labour.

In this economy, production is carried out using corn (capital) and labour, and the output is also corn. A technique of production, in this setting, is a combination of the quantities of corn and labour needed to produce 1 unit of corn, i.e., the combination of capital and labour productivity.

Initial Situation

Let the nominal wage rate be given by w , and let the technique of production be given by the combination (a, n) , i.e., a units of corn and n units of labour are needed to produce 1 unit of corn. Let the value (which is also equal to price) of a unit of corn be denoted by p . Then,

$$ap + n = p,$$

so that

$$p = \frac{n}{1-a}.$$

If we denote the real wage rate by b and the rate of profit by r , then we have

$$b = \frac{w}{p} = \frac{w(1-a)}{n}$$

and

$$r = \frac{p-(nw+ap)}{nw+ap} = \frac{1}{nb+a} - 1.$$

Innovator Capitalist

Suppose an innovator capitalist finds a new technique of production, given by the combination (a', n') . Thus, if the new technique of production is used, then a' units of corn and n' units of labour will be needed to produce 1 unit of corn. For this new technique of production to increase the rate of profit of the innovator, it must be "viable", i.e., it must reduce the cost of production at the prevailing prices, i.e., it must be the case that $wn + pa > wn' + pa'$, or equivalently that

$$bn' + a' < bn + a. \quad (A1)$$

Since we want to investigate conditions under which the new technique is adopted by the innovator capitalist, we will assume that (A1) holds. This will immediately imply that the rate of profit earned by the innovator capitalist with the new technique of production

$$r' = \frac{1}{n'b + a'} - 1$$

will be higher than what she earned using the old technique of production

$$r = \frac{1}{nb+a} - 1.$$

Since $r' > r$, there is an incentive for the innovator capitalist to search for an adopt the new technique of production.

The Marx-Okishio Threshold

When all capitalist firms adopt the new technique of production, the price (and value) of the commodity becomes

$$p' = \frac{n'}{1-a'}.$$

Suppose the new real wage rate is given by δb , so that the growth rate of the real wage rate is given by $(\delta - 1)$. In the new configuration of price and real wage rate, the rate of profit is given by

$$r'' = \frac{1}{a' + \delta b n'} - 1.$$

Let δ^* denote the value of δ that ensures that the average rate of profit before the adoption of the new technique of production, r , is exactly equal to the average rate of profit after the adoption of the new technique of production by all capitalist firms, r'' . Hence

$$\frac{1}{a' + \delta^* b n'} - 1 = \frac{1}{a + b n} - 1,$$

so that

$$\delta^* = \frac{n}{n'} + \left(\frac{a}{a'} - 1 \right) \frac{p a'}{w n'}.$$

The Marx-Okishio threshold, α^* , is the growth rate of real wage rate that ensures that ensures that the average rate of profit before the adoption of the new technique of production, r , is exactly equal to the average rate of profit after the adoption of the new technique of production by all capitalist firms, r'' . Hence,

$$\alpha^* = \delta^* - 1,$$

so that

$$\alpha^* = \left(\frac{n}{n'} - 1 \right) + \left(\frac{a}{a'} - 1 \right) \frac{p a'}{w n'}.$$

Note that

$$\left(\frac{n}{n'} - 1\right) = \frac{\left(\frac{1}{n'}\right) - \left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)}$$

is the growth rate of labour productivity, and

$$\left(\frac{a}{a'} - 1\right) = \frac{\left(\frac{1}{a'}\right) - \left(\frac{1}{a}\right)}{\left(\frac{1}{a}\right)}$$

is the growth rate of capital productivity. Moreover, $\frac{pa'}{wn'}$ is the new organic composition of capital (ratio of constant capital and variable capital) evaluated at prices prevailing before the adoption of the new technique of production. If we denote the growth rate of labour productivity by β_1 , the growth rate of capital productivity by β_2 , and the organic composition of capital (evaluated at prices prevailing before technical change) as γ , then we have the Marx-Okishio threshold as

$$\alpha^* = \beta_1 + \gamma\beta_2.$$

Now we are ready to prove the main results relating to the Marx-Okishio threshold.

Marx's Result

If the growth rate of the real wage rate is α , where $\alpha > \alpha^*$, then the average rate of profit will fall after the adoption of the new technique of production by all capitalists. To see this note that, since $\alpha > \alpha^*$, it is also true that

$$1 + \alpha > 1 + \alpha^* = \frac{n}{n'} + \left(\frac{a}{a'} - 1\right) \frac{pa'}{wn'} = \frac{n}{n'} + \left(\frac{a}{a'} - 1\right) \frac{a'}{bn'} = \frac{bn + (a - a')}{bn'}$$

which, on multiplying through by bn' gives the inequality

$$(1 + \alpha)bn' + a' > bn + a. \quad (A2)$$

The average rate of profit before the adoption of the new technique of production is given by

$$r = \frac{1}{bn+a} - 1.$$

On the other hand, if the growth rate of the real wage rate is α , then the average rate of profit after the adoption of the new technique of production is given by

$$\hat{r} = \frac{1}{(1+\alpha)bn' + a'} - 1.$$

Using the inequality in (A2), we see that $\hat{r} < r$.

Okishio's Result

If the growth rate of the real wage rate is α , where $\alpha < \alpha^*$, then the average rate of profit will rise after the adoption of the new technique of production by all capitalists. The proof follows in exactly the same manner as the proof for Marx's claim, with the sign of the key inequalities reversed.

A3. Expanded Reproduction in Capitalist Economies

To study the conditions for expanded reproduction, we follow Marx (1992) in viewing the whole of social production as being organized within two departments. Department I produces means of production and Department II produces means of consumption. The flow of time is broken into periods, with the following convention about activities: at the beginning of each period capital outlays are made; within the period production of commodities takes place; and at the end of the period the output is sold and the value (and surplus value) realized. Sale of the output comes from expenditures of capitalists and workers, the two classes in this economy.

The capitalist class makes two types of expenditures: (a) capital outlays to carry out production, which is the sum of constant capital (the amount of money used to purchase means of production) and variable capital (the amount of money used to purchase labour-power); and (b) consumption expenditure. Variable capital, which is part of the capital outlays by capitalists, corresponds to the wage income of the working class, which is used for the consumption expenditure of workers. Hence, the total expenditure by the capitalist class is the primary source of all demand in the model of capitalist economies that we study here (the most important exclusions are the State and the rest of the world, which can be an additional sources of autonomous expenditure).

We can re-group the expenditures in the economy to arrive at the sources of demand for the output of the two departments. The demand for the output of Department I come from two types of expenditures: (a) replacement of used-up means of production in Department I and II, and (b) increments to the means of production in Department I and II. Note that both expenditures arise directly from the capital outlays in the two departments.

The demand for the output of Department II also comes from two types of expenditures: (a) consumption expenditure of workers in Department I and II; and (b) consumption expenditures of capitalists in Department I and II. Consumption expenditure of workers comes from wage income, which is the variable capital component of capital outlays by capitalists. Thus, consumption expenditure of workers comes indirectly from the capital outlays of capitalists. On the other hand, consumption expenditure of capitalists comes from the surplus value realized through sale of the output.

For a complete description of the model, we need to specify three parameters related, in turn, to the technology of production, the degree of exploitation of workers, and the investment propensity of capitalists. To do so, we will use subscripts *I* and *II* to identify the two department. Let k_I and k_{II} denote the share of variable capital in total capital outlays in the two departments; these parameters characterize the technology of production used in the two departments, with a lower value of k denoting a more capital-intensive technology. Let e denote the rate of exploitation (ratio of surplus

value and variable capital) that is common in both departments. Let p_I and p_{II} refer to the share of surplus value reinvested into production in the two departments; these characterize the investment behavior of capitalists in the two departments, with a higher value of p signifying a higher propensity to save and invest. Note that if $p_I = p_{II} = 0$, then all the surplus value is consumed by capitalists and we get the special case of simple reproduction.

If this economy is to smoothly reproduce over time, it must be the case that there is neither excess demand nor excess supply in the market for the output of either department. Let us start by investigating the condition for equilibrium, i.e. when demand is exactly equal to supply, in the market for the output of Department I. Denoting constant capital by C , variable capital by V and surplus value by S , the total value of output in Department I is given by $C_I + V_I + S_I$. In equilibrium, this has to be equal to the total demand for the output of Department I. But what is the total demand for the output of Department I, i.e. means of production?

In Department I, the demand for means of production comes from the need to replace the means of production that has been used up, C_I , and the need for incrementing the means of production, ΔC_I . The increment to the means of production in Department I arises from the reinvestment out of surplus value. In Department I, $p_I S_I$ is the amount of surplus value reinvested, of which the fraction $(1 - k_I)$ is used for purchasing means of production. Hence, the increment to the means of production in department is given by $\Delta C_I = (1 - k_I)p_I S_I$. In a similar manner, the demand for means of production in Department II comes from replacement needs, C_{II} , and from the need to expand the means of production, ΔC_{II} , where the latter is given by the expression: $\Delta C_{II} = (1 - k_{II})p_{II} S_{II}$. Hence the total demand for means of production is given by

$$C_I + \Delta C_I + C_{II} + \Delta C_{II} = C_I + (1 - k_I)p_I S_I + C_{II} + (1 - k_{II})p_{II} S_{II}.$$

There is equilibrium in the market for means of production when the total supply is equal to the total demand, i.e.

$$C_I + V_I + S_I = C_I + (1 - k_I)p_I S_I + C_{II} + (1 - k_{II})p_{II} S_{II},$$

which simplifies to

$$V_I + S_I = (1 - k_I)p_I S_I + C_{II} + (1 - k_{II})p_{II} S_{II}.$$

Since $C_I = [(1 - k_I)/k_I]V_I$, $C_{II} = [(1 - k_{II})/k_{II}]V_{II}$, $S_I = eV_I$ and $S_{II} = eV_{II}$, we can plug these in the above equilibrium condition and algebraically manipulate it to get the following relation between the sizes of the two departments:

$$\frac{V_{II}}{V_I} = \frac{1 + e - ep_I(1 - k_I)}{\left[\frac{1 - k_{II}}{k_{II}}\right] + ep_{II}(1 - k_{II})}. \quad (A3)$$

The condition derived in (A3) would also emerge if we investigated the condition for equilibrium in the market for the output of Department II.²⁹ This is not surprising. Since total social production is broken up into two departments, equilibrium in the market for the output of one department will automatically imply equilibrium in the market for the output of the other.

Note that the analysis presented so far has implicitly ruled out an interesting possibility: the movement of capital between the two departments. This could happen, for instance, if some of the surplus value generated in Department I were invested in Department II in the next period, or vice versa. We have ruled this out by assuming that the investment to increase the scale of production in each department comes from surplus value generated within that department. This implicit assumption simplifies the analysis and allows us to get the following simple condition for balanced growth:

$$k_I p_I = k_{II} p_{II}. \quad (A4)$$

How do we get this condition given in (A4)? The intuition behind the condition is the following: if the two departments are to maintain the correct proportions over time without any inter-departmental transfer of capital, they will have to grow at the same rate *on the basis of the surplus value generated within the department*.

If growth in Department I comes from reinvestment of surplus value generated within the department *only*, then the rate of growth of variable capital in Department I is given by

$$g_I = \frac{\Delta V_I}{V_I} = \frac{k_I p_I S_I}{V_I} = \frac{k_I p_I e V_I}{V_I} = k_I p_I e.$$

Similarly, the rate of growth variable capital in Department II, on the basis of reinvestment of surplus value generated within the department only, is given by

$$g_{II} = k_{II} p_{II} e.$$

Hence, if

$$k_I p_I = k_{II} p_{II},$$

then both departments grow at the same rate and the correct proportions are maintained over time, which is the condition in (A4).

²⁹ The total supply of the output of Department II is $C_{II} + V_{II} + S_{II}$ and the total demand for the output of Department II is $V_I + \Delta V_I + V_{II} + \Delta V_{II} + (1 - p_I)S_I + (1 - p_{II})S_{II}$. Equating total supply with total demand, plugging in the correct expressions for the terms and manipulating them algebraically will give (A3).